

Assignment 4

Author: Jing Zhang

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Preliminaries

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 3.2.5
```

```
library(rgeos)
```

```
## Warning: package 'rgeos' was built under R version 3.2.5
```

```
## rgeos version: 0.3-22, (SVN revision 544)
## GEOS runtime version: 3.4.2-CAPI-1.8.2 r3921
## Linking to sp version: 1.2-4
## Polygon checking: TRUE
```

```
library(haven)
```

```
## Warning: package 'haven' was built under R version 3.2.5
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.2.5
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:rgeos':
```

```
##
```

```
## intersect, setdiff, union
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(jsonlite)
```

```
## Warning: package 'jsonlite' was built under R version 3.2.5
```

```
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 3.2.5
```

```
library(stringr)
```

Assignment Part I

1) *Answer to Problem 1*

$$\begin{aligned} & \int (2x^3 + 3x - 1) dx \\ &= \frac{1}{2}x^4 + \frac{3}{2}x^2 - x + C \end{aligned}$$

2) *Answer to Problem 2*

$$\begin{aligned} & \int_0^1 \frac{2}{x+3} dx \\ &= 2 \ln(1+3) - 2 \ln(0+3) \\ &= 2 \ln\left(\frac{4}{3}\right) \end{aligned}$$

Range of integration $[-5, -4]$: the limits are on the same side of -3 and the integral is $-2 \ln(2)$

$$\begin{aligned} \int_{-5}^{-4} \frac{2}{x+3} dx &= 2 (\ln |x+3|) \Big|_{-5}^{-4} \\ &= 2 \ln 1 - 2 \ln 2 \\ &= -2 \ln 2. \end{aligned}$$

Range of integration $[-4, -2]$: the limits are on opposite side of -3 , so the integral is not defined.

3) *Answer to Problem 3*

In each part, denote the vectors by **a**, **b**, **c**

c) Suppose there are scalars α, β, γ such that

$$\begin{aligned}\alpha + 4\beta + 7\gamma &= 0 \\ 2\alpha + 5\beta + 8\gamma &= 0 \\ 3\alpha + 6\beta + 9\gamma &= 0\end{aligned}$$

Elimination steps lead to

$$\begin{aligned}\alpha &= \gamma \\ \beta &= -2\gamma\end{aligned}$$

So the vectors are linear dependent. We have $\mathbf{a}-2\mathbf{b}+\mathbf{c}=\mathbf{0}$

d) Suppose there are scalars α, β, γ such that

$$\begin{aligned}\alpha + 2\beta + 2\gamma &= 0 \\ 2\alpha - 2\beta + \gamma &= 0 \\ \alpha - \beta + 3\gamma &= 0\end{aligned}$$

Elimination steps lead to

$$\begin{aligned}\alpha + 2\beta + 2\gamma &= 0 \\ \beta - \frac{1}{3}\gamma &= 0 \\ \frac{5}{2}\gamma &= 0\end{aligned}$$

Back-substitution then gives $\alpha = \beta = \gamma = 0$, so that vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are linearly independent.

f) Suppose there are scalars α, β, γ such that

$$\begin{aligned}\alpha + 2\beta + 2\gamma &= 0 \\ 2\alpha - 2\beta + \gamma &= 0 \\ \alpha + 4\beta + 3\gamma &= 0\end{aligned}$$

Elimination steps lead to

$$\begin{aligned}\alpha + 2\beta + 2\gamma &= 0 \\ 2\beta + \gamma &= 0\end{aligned}$$

and the third is 0. Ignoring this uninformative third equation we have two equations in the three unknowns α, β, γ . We could find the complete solution by assigning an arbitrary value to γ and then find α and β in terms of it. Simply putting $\gamma=-2$ lead to $\beta=1$ and $\alpha=2$, such that $2\mathbf{a}+\mathbf{b}-2\mathbf{c}=\mathbf{0}$

4) *Answer to Problem 4*

We want to solve $\mathbf{Ax}=0$ for $\mathbf{x}$, with

$$\begin{bmatrix} 1 & -3 & 1 \\ 2 & -4 & 4 \end{bmatrix} \quad (3)$$

We start with augmented matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -3 & 1 \\ 2 & -4 & 4 \end{bmatrix} \quad (3)$$

Using the top row as pivot we multiply it by 2 and subtract from the second, we are left with:

$$\begin{bmatrix} 1 & -3 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{bmatrix} \quad (3)$$

We have 2 equations and 3 unknowns; assign arbitrary values to $x_3 = \mu$

$$\begin{aligned} 2x_2 + 2\mu &= 0 \\ x_2 &= -\mu \end{aligned}$$

and

$$\begin{aligned} x_1 + 3\mu + \mu &= 0 \\ x_1 &= -4\mu \end{aligned}$$

Simply putting $\mu=1$ lead to $x_1=-4$ and $x_2=-1$, such that

$$\begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix} \quad (3)$$

is a solution.

Also, simply putting $\mu=0$ lead to $x_1=0$ and $x_2=0$, such that $\mathbf{x}=\mathbf{0}$ is also a solution.

5) *Answer to Problem 5*

Since

$$\begin{bmatrix} x \\ -y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

the matrix describing the notation is

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (3)$$

6) *Answer to Problem 6*

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 1 & -2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 1 + 2 \times 2 & 1 \times 0 + 2 \times 1 & 1 \times (-1) + 2 \times 4 \\ 2 \times 1 + 0 \times 2 & 2 \times 0 + 0 \times 1 & 2 \times (-1) + 0 \times 4 \\ 1 \times 1 - 2 \times 2 & 1 \times 0 - 2 \times 1 & 1 \times (-1) - 2 \times 4 \\ 5 \times 1 - 2 \times 2 & 5 \times 0 - 2 \times 1 & 5 \times (-1) - 2 \times 4 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 5 & 2 & 7 \\ 2 & 0 & -2 \\ -3 & -2 & -9 \\ 1 & -2 & -13 \end{bmatrix}$$

$$\begin{aligned} \mathbf{AC} &= \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 1 & -2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 4 & -4 & 3 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 4 + 2 \times 1 & 1 \times (-4) + 2 \times 0 & 1 \times 3 + 2 \times (-1) & 1 \times 1 + 2 \times 0 \\ 2 \times 4 + 0 \times 1 & 2 \times (-4) + 0 \times 0 & 2 \times 3 + 0 \times (-1) & 2 \times 1 + 0 \times 0 \\ 1 \times 4 - 2 \times 1 & 1 \times (-4) - 2 \times 0 & 1 \times 3 - 2 \times (-1) & 1 \times 1 - 2 \times 0 \\ 5 \times 4 - 2 \times 1 & 5 \times (-4) - 2 \times 0 & 5 \times 3 - 2 \times (-1) & 5 \times 1 - 2 \times 0 \end{bmatrix} \\ &= \begin{bmatrix} 6 & -4 & 1 & 1 \\ 8 & -8 & 6 & 2 \\ 2 & -4 & 5 & 1 \\ 18 & -20 & 17 & 5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{CA} &= \begin{bmatrix} 4 & -4 & 3 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 1 & -2 \\ 5 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 4 \times 1 - 4 \times 2 + 3 \times 1 + 1 \times 5 & 4 \times 2 - 4 \times 0 + 3 \times (-2) + 1 \times (-2) \\ 1 \times 1 - 0 \times 2 - 1 \times 1 + 0 \times 5 & 1 \times 2 + 0 \times 0 - 1 \times (-2) + 0 \times (-2) \end{bmatrix} \\ &= \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \end{aligned}$$

Assignment Part II

7) *Answer to Problem 7*

The dta1 is tidy. In this dataset, each column is a variable and each row is an observation. With Country A, Country B and year, we can uniquely identify the observation.

The dta2 is tidy. With year, Dyad and conflictType, we can uniquely identify the observation.

The dta3 is not tidy. Two variables are clumped together in one column.

```
dta1<- data.frame(year = c(2000:2003,2000:2003),
  CountryA = c(rep("USA",4),rep("CAN",4) ),
  CountryB=c(rep("Russia",4),rep("Russia",4) ),
  Agreement=c(0,1,1,0,0,1,0,0))
```

```
dta2<- data.frame(year = c(2000:2003,2000:2003),
  Dyad = c(rep("USA-RUS",4),rep("USA-RUS",4) ),
  ConflictType=c(rep("War",4),rep("Minor MID",4) ),
  ConflictLevel=c(0,3,1,0,0,1,0,0))
```

```
dta3<- data.frame(Dyad = c("USA-RUS", "CAN-RUS"),
  War2002=c(0,2),
  War2003=c(0,2),
  War2004=c(0,2),
  MinorMID2002=c(0,2),
  MinorMID2003=c(0,2),
  MinorMID2004=c(0,2))
```

```
head(dta1)
```

```
##   year CountryA CountryB Agreement
## 1 2000      USA   Russia         0
## 2 2001      USA   Russia         1
## 3 2002      USA   Russia         1
## 4 2003      USA   Russia         0
## 5 2000      CAN   Russia         0
## 6 2001      CAN   Russia         1
```

```
head(dta2)
```

```
##   year   Dyad ConflictType ConflictLevel
## 1 2000 USA-RUS          War             0
## 2 2001 USA-RUS          War             3
## 3 2002 USA-RUS          War             1
## 4 2003 USA-RUS          War             0
## 5 2000 USA-RUS   Minor MID             0
## 6 2001 USA-RUS   Minor MID             1
```

```
head(dta3)
```

```
##           Dyad War2002 War2003 War2004 MinorMID2002 MinorMID2003 MinorMID2004
## 1 USA-RUS         0         0         0             0             0             0
## 2 CAN-RUS         2         2         2             2             2             2
```

8) *Answer to Problem 8*

```
fixednames<-c("Dyad","War-2002","War-2003","War-2004","MinorMID-2002","MinorMID-2003","MinorMID-2004")
names(dta3)<-fixednames
```

```
dta3tidy<-dta3 %>%gather(ConflictType, ConflictNumber,-Dyad)%>%separate(ConflictType, c('ConflictType',
dta3tidy
```

```
##           Dyad ConflictType Year ConflictNumber
## 1 USA-RUS          War 2002             0
## 2 CAN-RUS          War 2002             2
## 3 USA-RUS          War 2003             0
## 4 CAN-RUS          War 2003             2
## 5 USA-RUS          War 2004             0
## 6 CAN-RUS          War 2004             2
## 7 USA-RUS   MinorMID 2002             0
## 8 CAN-RUS   MinorMID 2002             2
## 9 USA-RUS   MinorMID 2003             0
## 10 CAN-RUS   MinorMID 2003             2
## 11 USA-RUS   MinorMID 2004             0
## 12 CAN-RUS   MinorMID 2004             2
```

9) *Answer to Problem 9*

a)

```
mean(c(3000, 2, 1, 0, 4)) == 601.4
```

```
## [1] TRUE
```

b)

```
EuclidDistancetoKabul <- function(x) {  
  sqrt((-64 - x[1]) ^ 2 + (16 - x[2]) ^ 2)  
}  
v<-rbind(c(48, 4), c(12, 3), c(80, 13))  
x<-apply(v, 1, EuclidDistancetoKabul) # 1 row-wise, 2 column-wise  
x > 20
```

```
## [1] TRUE TRUE TRUE
```

c)

```
install.packages("car", repos="http://cran.rstudio.com/")
```

```
## Warning: dependency 'pbkrtest' is not available
```

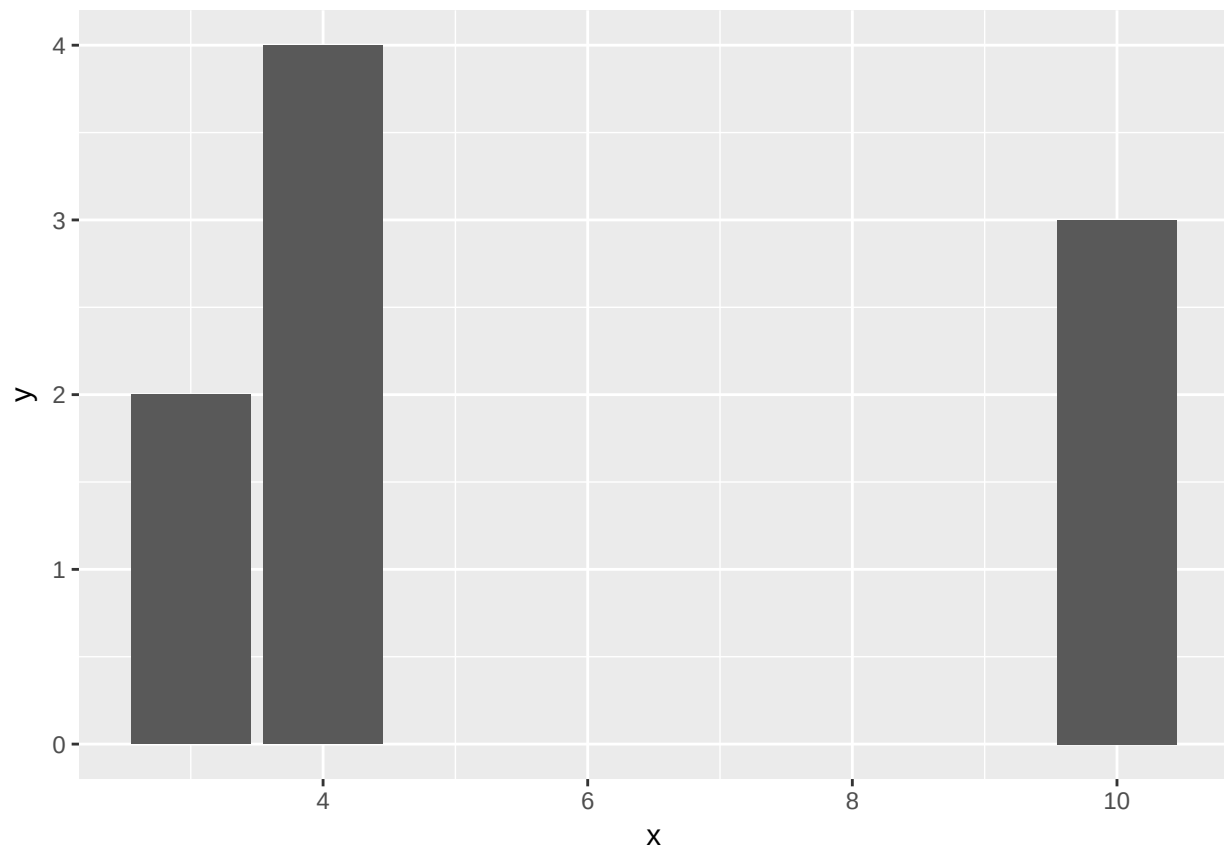
```
##
```

```
## The downloaded binary packages are in
```

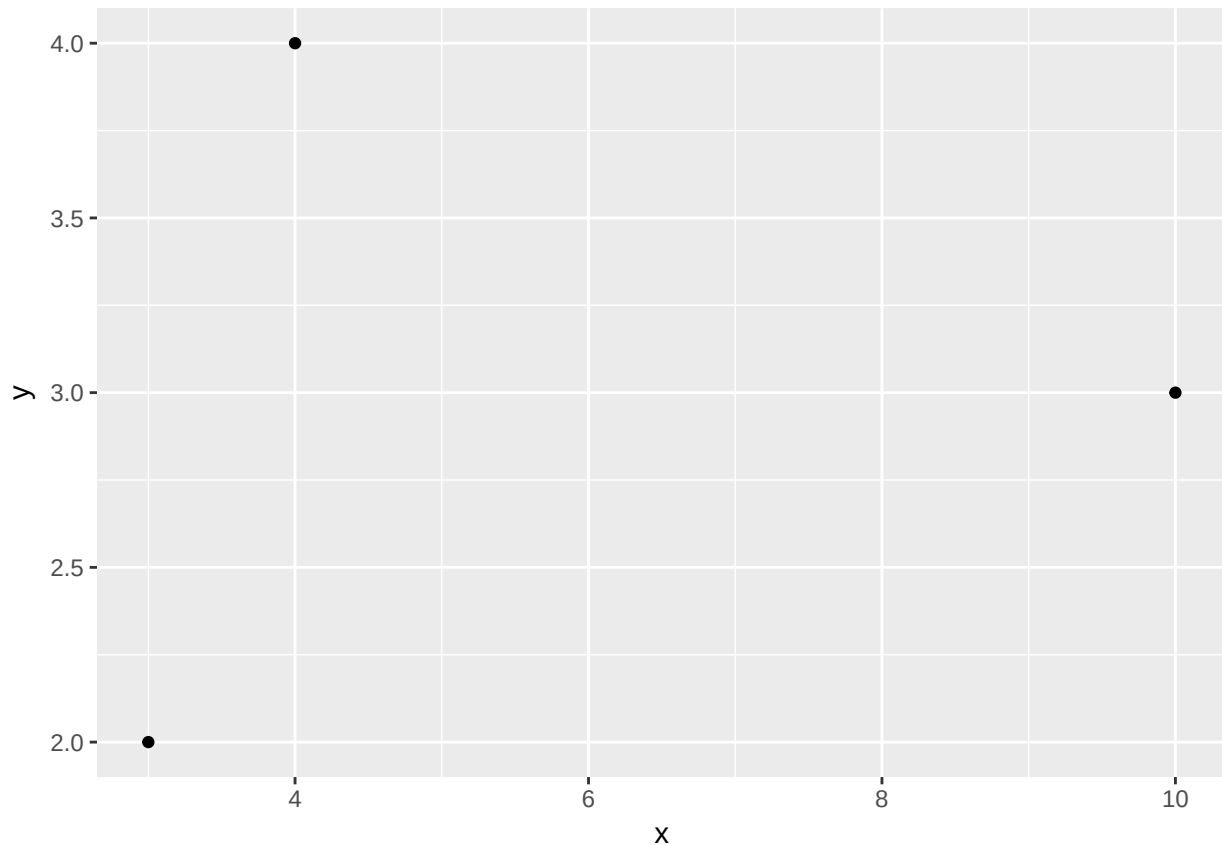
```
## /var/folders/_m/_pytmsvd1t16mbk79wkx275r0000gn/T//RtmpE3JLgl/downloaded_packages
```

d)

```
x <- c(3, 10, 4)  
y <- c(2, 3, 4)  
df <- data.frame(x, y)  
ggplot(df, aes(x,y)) + geom_bar(stat="identity") #The column you are trying to summarize with geom_bar()
```



```
x <- c(3, 10, 4)
y <- c(2, 3, 4)
df <- data.frame(x, y)
ggplot(df, aes(x, y)) +
  geom_point()
```

e)

```
allthetypes <- data.frame(x = c("Iowa"), stringsAsFactors = F)
is.character(allthetypes[1,1])
```

```
## [1] TRUE
```

10) *Answer to Problem 10*

I define VA hospital activity to be the number of Total Service Users. 120 shootings happen in the state with most active VA hospital, which is Fl. 8 shootings happen in the state with least active VA hospital, which is Wyoming. (The state with least active VA hospital is PI, but there was no information about shootings there).

```
VAest <- fromJSON(paste0("https://raw.githubusercontent.com/vacobrydsk/",
                        "VHA-Files/master/NEPEC_AnnualDataSheet_MH_FY15.json"), flatten = T)
```

```
URLchunkA <- "https://raw.githubusercontent.com/washingtonpost/"
URLchunkB <- "data-police-shootings/master/fatal-police-shootings-data.csv"
shootings <- read.csv(paste0(URLchunkA, URLchunkB))
```

```
VAtidy <- VAest %>% select(-ValueType, -Category) %>% spread(Item, Value)
```

```
fixednames2 <- c("Station", "ADC_all_beds", "ADC_inpatient", "ADC_residential",
                 "ADC_psych_encounters", "ADC_psych_unique", "Avestops_MH",
```

```

      "Avestops_Psychiatry", "Avestops_Substance", "Setting_Inpatient",
      "Setting_Outpatient", "Setting_Resident", "N_serviceusers",
      "N_serviceusers_confirmed", "N_serviceusers_possible", "Pct_confirmed",
      "Pct_possible", "NMH_Prop_community_confirmed", "NMH_Prop_inpatient_confirmed",
      "NMH_Prop_outpatient_confirmed", "NMH_Prop_community_possible",
      "NMH_Prop_inpatient_possible", "NMH_Prop_outpatient_possible",
      "Prop_Vets_confirmed_any", "Prop_Vets_confirmed_outpatient",
      "Prop_Vets_confirmed_inpatient", "Prop_Vets_confirmed_resident",
      "Prop_Vets_possible_any", "Prop_Vets_possible_outpatient",
      "Prop_Vets_possible_inpatient", "Prop_Vets_possible_resident",
      "testing_encounters", "testing_uniques", "ServiceUserswithMH",
      "Total_NHM_stays_community", "Total_NHM_stays_inpatient",
      "Total_NHM_stays_outpatient", "Total_Service_Users", "Unique_Vets_inpatient",
      "Unique_Vets_outpatient", "Unique_Vets_resident")
names(VAtidy) <- fixednames2

```

```

stringtonumber <- function(sortanumber) {as.numeric(str_replace_all(as.character(sortanumber), "%|,", ""))}

VAtidy2 <- VAtidy %>% mutate_all(stringtonumber)

```

```

## Warning in `__dplyr_colwise_fun`(c("358", "402", "405", "436", "437",
## "438", : NAs introduced by coercion

```

```

VAFacilities <- fromJSON(paste0("https://raw.githubusercontent.com/department-of-veterans-affairs",
                                "/VHA-Facilities/master/VAFacilityLocation.json"), flatten = T)
VAFacdata <- VAFacilities$VAFacilityData %>% mutate(Sttn = facility_id)

```

```

library(dplyr)
VAtidy2<- VAtidy2 %>% select(Sttn= Station, `Total Service Users` = `Total_Service_Users`)
VAFacdata2 <- merge(VAFacdata, VAtidy2, by="Sttn")
shootingbystate <- shootings %>% group_by(state) %>% summarize (nshot= length(race))
facilitywshooting <- merge (shootingbystate, VAFacdata2, by="state")

facilitywshooting2 <- facilitywshooting[order(-facilitywshooting$`Total Service Users`), ]
facilitywshooting2[1:3,1:2, drop= TRUE]

```

```

##      state nshot
## 60     FL    122
## 67     FL    122
## 75     GA     60

```

```

facilitywshooting3 <- facilitywshooting[order(facilitywshooting$`Total Service Users`), ]
facilitywshooting3[1:4,1:2, drop= TRUE]

```

```

##      state nshot
## 323     WY     8
## 326     WY     8
## 328     WY     8
## 329     WY     8

```

11)

```
A <- function(x) {  
  y <- (1/sqrt(2*pi))*(exp((-x^2)/2))  
  return(y)  
}  
B <- function(llim,ulim) {  
  x <- seq(-100,100,0.01)  
  y <- A(x)  
  x_lim <- seq(llim,ulim,0.05)  
  plot(x,y, cex= 0.1, xlim = c(-5,5))  
  for (i in 1:length(x)){  
    coord.x <- c(x_lim[i],x_lim[i], x_lim[i+1],x_lim[i+1])  
    coord.y <- c(0,A(x_lim[i]), A(x_lim[i+1]),0)  
    polygon(coord.x,coord.y,col='gray', border=NA)  
  } }  
B(-4,4)
```

